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10th International Mathematics Assessments for Schools (2020-2021)

Middle Primary Division Round 2

Time: 120 minutes

Printed Name _____ *Code* _____ *Score* _____

Instructions:

- Do not open the contest booklet until you are told to do so.
- Be sure that your name and code are written on the space provided above.
- Round 2 of IMAS is composed of three parts; the total score is 100 marks.
- Questions 1 to 5 are given as a multiple-choice test. Each question has five possible options marked as A, B, C, D and E. Only one of these options is correct. After making your choice, fill in the appropriate letter in the space provided. Each correct answer is worth 4 marks. There is no penalty for an incorrect answer.
- Questions 6 to 13 are a short answer test. Only Arabic numerals are accepted; using other written text will not be honored or credited. Some questions have more than one answer, as such all answers are required to be written down in the space provided to obtain full marks. Each correct answer is worth 5 marks. There is no penalty for incorrect answers.
- Questions 14 and 15 require a detailed solution or process in which 20 marks are to be awarded to a completely written solution. Partial marks may be given to an incomplete presentation. There is no penalty for an incorrect answer.
- Use of electronic computing devices is not allowed.
- Only pencil, blue or black ball-pens may be used to write your solution or answer.
- Diagrams are not drawn to scale. They are intended as aids only.
- After the contest the invigilator will collect the contest paper.

The following area is to be filled in by the judges;
the contestants are not supposed to mark anything here.

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	Total Score	Signature
Score																	
Score																	

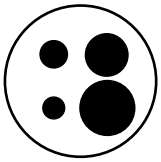
Middle Primary Division Round 2

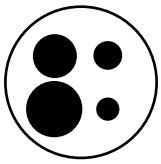
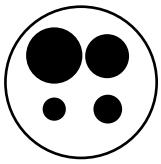
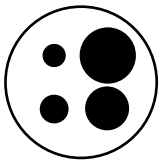
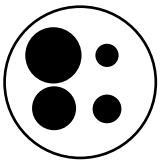
Questions 1 to 5, 4 marks each

1. Which of the following expressions has a value equal to 3?
（A）$2\times0+2-1$　　（B）$2-0+2\times1$　　（C）$2+0\times2+1$
（D）$2+0+2+1$　　（E）$2\times0+2\div1$

Answer：_____

2. Which among the choices has the same pattern as the figure shown below? (Note: figure can be rotated but not flipped)



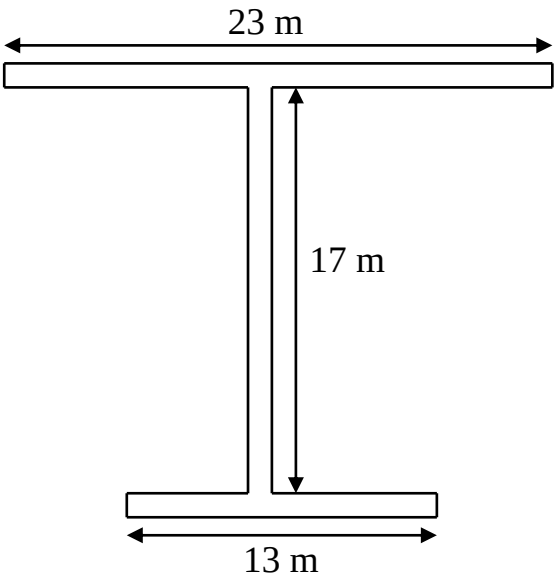
（A） 
（B） 
（C） 
（D） 
（E） 

Answer：_____

3. Nelson has some number of apples and one pear that he must place in a straight line on the table. If we start counting from left to right, the pear is located on the 12th position, while counting from right to left, the pear is located on the 18th position. How many fruits are there on the table?
（A）30　　（B）29　　（C）28　　（D）22　　（E）20

Answer：_____

4. Jack paved the sidewalk in the garden with tiles as shown in the figure below. If the width of the sidewalk is 1 m, what is the total area, in m^2, of the sidewalk?



（A）53　（B）104　（C）106　（D）221　（E）391

Answer：_____

5. What is the value of the expression $1234+2341+3412+4123$?

（A）5555　（B）10000　（C）11110　（D）11111　（E）10101010

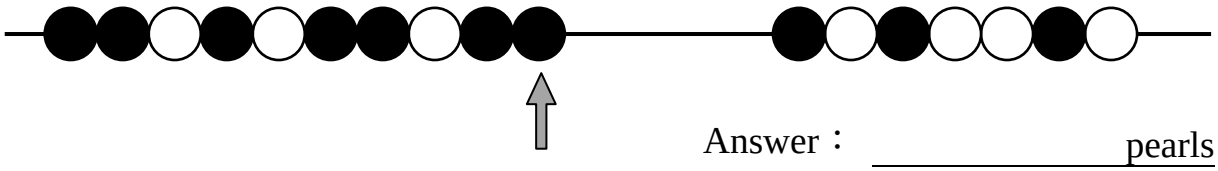
Answer：_____

Questions 6 to 13, 5 marks each

6. Using the dd/mm/yyyy (day/month/year) format, dates such as 01/02/2010 and 02/02/2020 are called *"Palindromic Days"* because the eight-digit numbers formed (e.g. 01022010 and 02022020) are both palindromes. What is the remainder when the *"Palindromic Day"* in the year 2021 is divided by 11?
(Note: A palindrome is a number which reads the same backward as forward.)

Answer：_____

7. In the figure below, we use a thread to string some black and white pearls into a straight line. Starting from the black pearl marked with an arrow, we move the pearls one by one from left to right. How many pearls must be moved so that the number of black pearls on the left is equal to the number of white pearls on the right?

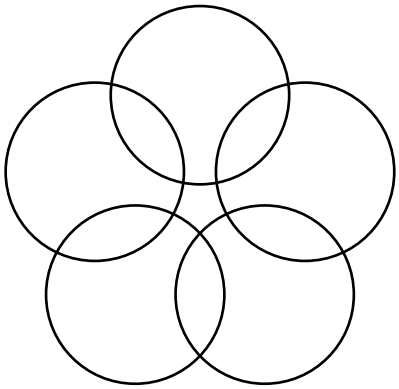


Answer：_____ pearls

8. There are 3 air conditioning units in John's house and one new air conditioning unit was installed over the summer. However, due to power limitations of his electric meter, if all 4 air conditioners are turned on at the same time, the fuse will overload, so only 3 of them can be turned on at the same time. So, in a 24 hour period over the summer, if 3 air-conditioning units are always turned on, then at least how long is the most used air conditioning unit turned on?

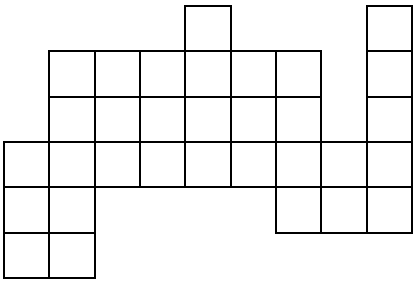
Answer： _____ hours

9. The following figure shows five circles that intersect each other in pairs. If each circle has an area of 6 m^2 and each overlapping area of two intersecting circles is 1 m^2, then what is the total area, in m^2, covered by the five circles?



Answer： _____ m^2

10. The following figure is made up of some number of identical unit squares. Now, Alex wants to form a bigger square by attaching unit squares on the figure (without changing the current figure). At least how many unit squares must he use in order to do so?



Answer： _____

11. Using some number of black and white balls of the same size, we arrange the figures as shown in the diagram below. If Figure 1 uses 8 balls, Figure 2 uses 13 balls, Figure 3 uses 20 balls and so on. How many balls are there in Figure 10?

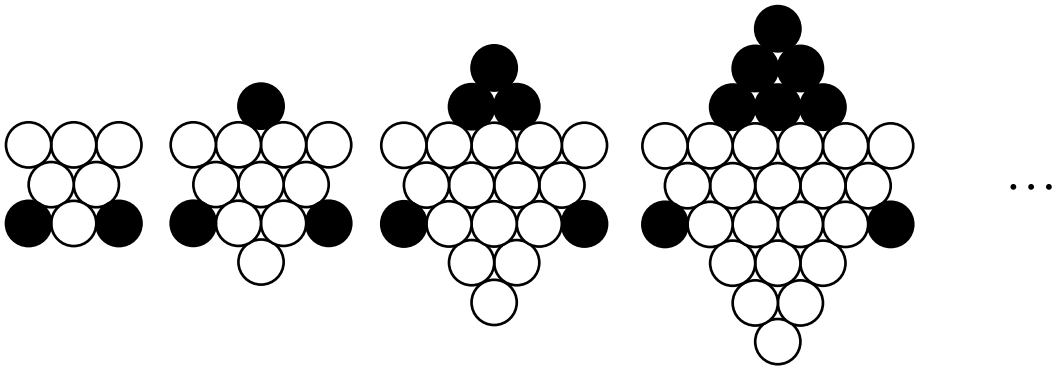
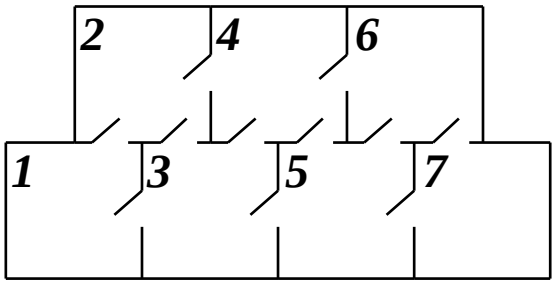


Figure 1 Figure 2 Figure 3 Figure 4

Answer : _____ balls

12. The diagram below shows seven rooms that are numbered from 1 to 7 such that each room has a door connected to any of its adjacent rooms. Starting from room 1, our goal is to reach room 7, by passing through other rooms. If you can only move to a room with the larger number, then how many different routes are there?



Answer : _____ routes

13. Fill in the letters *a*, *b*, *c*, *d* and *e* in each of the cells of the 5×5 table shown below such that each letter appears exactly once in each row and column. What letter should be placed in the cell where the "☆" is located?

a	*b*			
				a
	☆	*d*		
b		*e*		
	e			*d*

Answer : _____

Questions 14 to 15, 20 marks each
(Detailed solutions are needed for these two problems)

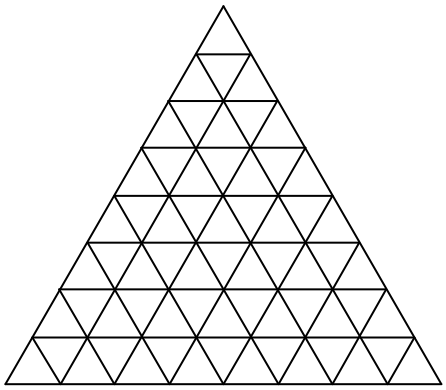
14. If 2021 is expressed as a sum of 100 prime numbers, where the largest prime number used must be as small as possible, then what is the smallest possible value of the largest prime number among these 100 prime numbers? Give an example that satisfies these conditions.

Answer： The smallest possible value of the largest prime number is .

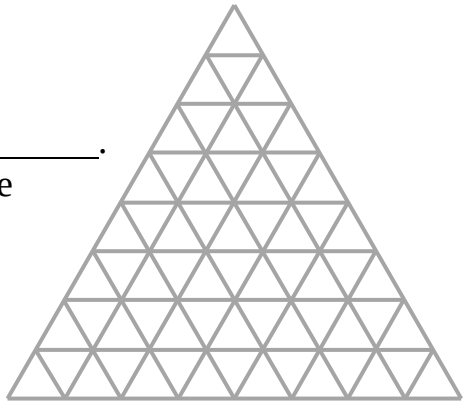
We can choose the following primes to satisfy the conditions:

15. The two figures below are composed of small equilateral triangles of the same size. We want to put several pieces of the parallelogram cardboard (which can be rotated or flipped that is shown on the left) into the equilateral triangle grid that is shown on the right without overlapping each other. The sides of these parallelogram cardboards are required to be placed along the grid lines of the equilateral triangle grid and cannot exceed the sides of it. What is the maximum number of pieces of parallelogram cardboard that can be placed into the triangular grid?
Justify your answer and show one way to place the cardboards.





The maximum number of pieces is _____.
Right figure is one way to place those cardboards.



Answer：_____